François Rose* and James E. Hetrick†
*Conservatory of Music
†Physics Department
University of the Pacific
3601 Pacific Avenue
Stockton, California 95211 USA
{frose, jhetrick}@pacific.edu

Enhancing Orchestration Technique via Spectrally Based Linear Algebra Methods

Spectral analysis has a rich and central place in the history and understanding of music, and modern music production relies heavily upon it. However, the advanced use of Fourier techniques as an aid to composition—as opposed to a tool for signal processing or sound synthesis—has been relatively unexplored until recently (Carpentier et al. 2006, 2007; Carpentier, Tardieu, and Rodet 2007; Hummel 2005; Psenicka 2003; Rose and Hetrick 2005, 2007).

Because orchestration is the vehicle that carries a musical idea from imagination to reality, it is clear that composers' orchestration techniques have a major impact on their musical expressions. Spectral analysis, by providing essential information about sound mixtures that are new or that are subject to constraints, can profoundly and positively influence that technique.

In this article, we present our approach to such a computerized aid that extends the use of spectral analysis for orchestration. It uses a bank of Discrete Fourier Transforms (DFTs) of orchestral sounds, which are accessed in different ways designed to either perform sound analysis or propose orchestrations that imitate the energetic pattern of a reference sound. The output of the tool is a weighted set of all possible sound mixtures from the *palette*, a subset of the complete bank—typically instruments for which the composer is writing.

Although our tool (Linear Algebra Based ORCHestration) is designed as an aid for composers, it is not our intent to propose a replacement of traditional orchestration technique with the blind use of new technology. After all, the sophisticated level reached by the empirical development of orchestration technique is proof that imagination and intuition are the composer's most valuable and irreplaceable assets. Rather, we are proposing to combine the two, because we firmly believe that the integration of spectral analysis in orchestration technique can expand the boundaries of the composer's imagination and intuitive flair.

The article is organized as follows. After reviewing related research, we introduce the structure and mathematics of our tool and its potential to analyze and compare different orchestrations. Then, the tool's main asset, its capability to produce orchestral propositions, is illustrated with an example from the works of the first author: *L'identité voilée*. Finally, we sketch unresolved issues and the directions we intend to follow with our approach.

Computer Music Journal, 33:1, pp. 32–41, Spring 2009
© 2009 Massachusetts Institute of Technology.

Related Research

Carpentier and colleagues (Carpentier et al. 2006, 2007; Carpentier, Tardieu, and Rodet 2007) introduced a tool that generates orchestral mixtures that imitate a target sound. The tool uses a shorthand instrument database, reduced from a larger sound-sample database. A sound is analyzed and defined in the instrument database in terms of acoustical features: pitch, fundamental frequency, spectral centroid, and harmonic spectrum, each of which affects the timbre. Based on pitch criteria, the orchestration engine selects sounds from the instrument database that could potentially be part of the proposed orchestral mixtures. Then, using a hybrid genetic/local-search algorithm, combinations are created. These are then evaluated using a multi-objective approach, searching for the sound mixture that has the best value for each of these acoustical features.

Hummel (2005) proposes a method that synthesizes the spectral envelope of a phoneme using the spectral envelopes of different sounds of musical instruments. A database of spectral envelopes is created by measuring, for each sound, its absolute amplitude in dBA, then calculating and normalizing its spectral envelope. The program then computes the target's spectral envelope and iteratively accesses

the database to find the best approximation. First, the spectral envelope of an instrument that best matches the target's spectral envelope is identified. Subtracting this envelope from the target's creates a new envelope. The tool then searches for the best match to this new envelope. The process is repeated until the target's spectral envelope is simulated sufficiently well.

Psenicka (2003) presents a similar method in his Lisp-based program called SPORCH (SPectral ORCHestration), but his iterative matching algorithm works on spectral peaks rather than spectral envelopes. In this case, each instrument is defined in the reduced database in terms of its most prominent peak values taken from an analysis of each pitch at various dynamic levels. The program then computes the target's spectral peaks and iteratively accesses the database to find the best approximation. A match is made if the two peaks occur within a certain specified frequency range, in which case the amplitude value of the instrument peak is subtracted from the amplitude of the matching target's peak. Another match is undertaken using the amended target, and so on. The iteration continues until either no instrument is found that decreases the target data or all of the instrument combinations have been used.

Like our research, these approaches use the DFT to represent sounds. However, their algorithms manipulate and compare a further-reduced representation of the DFT, whereas our approach considers the full DFT in each of our algorithms.

Structure of Our Tool
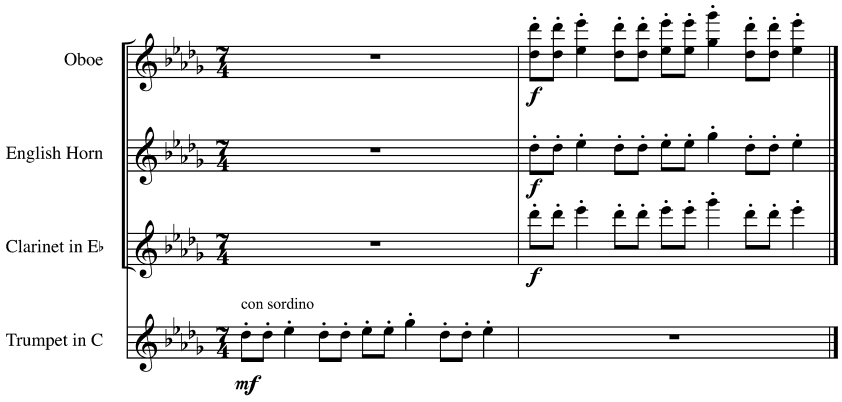
The tool that we propose works in conjunction with a bank of DFTs. Although we recognize that timbre is a psychoacoustic quality, its representation using the DFT is adequate for our purposes. To be a useful compositional aid, we felt that the tool had to be able to perform rather fast spectral analysis on different sound mixtures. That is why we decided that the most practical solution was to work directly with a bank of DFTs as opposed to sound samples themselves.

To ensure an acceptable degree of reliability, the bank was built in the following way. Sounds were all recorded under the same conditions, sampling at 44.1 kHz. Starting from the beginning of the sustain portion of each sound, a Hamming window of 4,096 samples was Fourier-transformed, and the norm of the complex DFT was stored. We then incremented by 512 samples and gathered another 4,096 samples, transformed them, and so on, until the end of the sustain portion was reached. The hundreds of DFTs generated by this process were then root mean square (RMS)-averaged into a single one; consequently, a bank sound is summarized by a single averaged DFT of length 4,096. Each averaged DFT was then compared with the two averaged DFTs adjacent to it in the chromatic scale. If the energy pattern of an averaged DFT did not logically fit between the adjacent ones, the averaged DFT was rejected, and another recording of that pitch was analyzed.

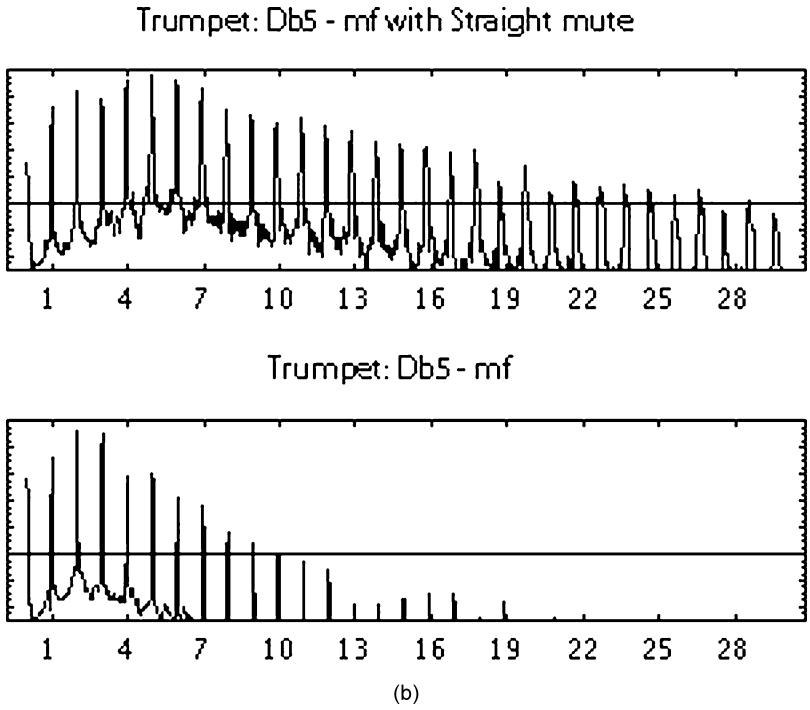
All pitches and performance techniques playable on an instrument are recorded at three different dynamic levels: *pp*, *mf*, and *ff*. One of the advantages of working with a bank of DFTs is that the spectral envelope of any mixture of instruments from the bank can be rapidly produced. Indeed, the tool uses the standard linear combination of DFTs to generate these spectral envelopes.

With our bank of DFTs alone, we are already in a position to make some interesting observations regarding orchestration. An example from the opening of *La danse des jeunes filles* in Ravel's *Daphnis and Chloé* is used to illustrate how simply examining sounds in the bank can assist composers in their orchestration decisions. In *Daphnis et Chloé*, Ravel uses a fascinating counterpoint of timbres at the beginning of *La danse des jeunes filles*, at rehearsal number 17. In the foreground, a solo muted trumpet is answered one measure later by the second oboe and English horn in octave doubling, with the first oboe and the E-flat clarinet, as shown in Figure 1a. (The transcription is in concert pitch.) It seems clear that Ravel is using the woodwinds to imitate the sound of the muted trumpet. We have used the tool to analyze the sound of the woodwind instruments and to determine which trumpet mute would be best suited to imitate it. The indication *con sordino* in a trumpet's part is almost always read as "straight mute on." Figure 1b shows two average spectra. The top is an averaged spectrum of a trumpet playing a D-flat5 *mezzo-forte* (554 Hz), muted with a straight mute. The bottom

Figure 1. (a) Foreground instruments of the opening two measures of Ravel's La danse des jeunes filles, *from* Daphnis et Chloé. *(b) Averaged spectra of a trumpet muted with a straight mute, and unmuted trumpet, both playing a D-flat5* mf. *(c) Top: averaged spectrum of a trumpet muted with a straight mute playing a D-flat5* mf*; middle: mix of the second oboe and an English horn playing a D-flat5* f *and the first oboe and an E-flat clarinet playing a D-flat6* f*; bottom: average spectrum of a trumpet muted with a cup mute playing a D-flat5* mf.



(a)

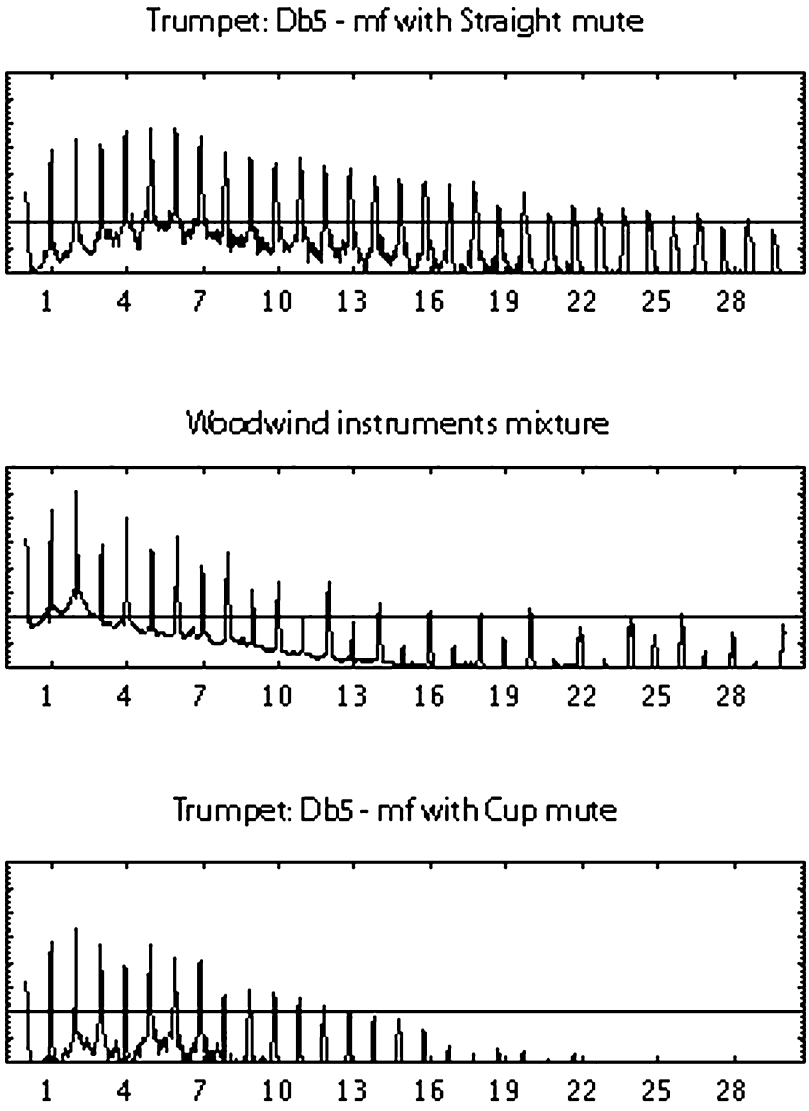


(b)

graphic is an averaged spectrum of an unmuted trumpet playing the same pitch and dynamic. A comparison of these two spectra shows that the straight mute behaves like a high-pass filter.

Figure 1c shows three averaged spectra. The top is again the spectrum of a trumpet playing a D-flat5 *mezzo-forte*, muted with a straight mute, and the middle spectrum shows the mixture of the four woodwind instruments. Note that the main energy is located in its lower part, more specifically around its second partial (D-flat6; 1,109 Hz), and that there is a cut-off around the 16th partial. This spectrum does not correspond very well with the one using the straight mute, which has its main energy around the 5th partial (F7; 2,794 Hz) and has an extremely rich structure, including more than 25 partials. On the other hand, the last spectrum in Figure 1c displays the sound of a trumpet muted with a cup mute, playing a D-flat5 *mezzo-forte*. Note that in this case, the main energy is around the second partial, and that there is a cut-off around the 12th partial. Therefore, we suggest that the resemblance between

Figure 1. Continued.



(c)

the trumpet's sound and the four woodwind instruments would be enhanced if the trumpet were to be muted with a cup mute instead of a straight mute.

Orchestration Propositions

Sounds are represented in the bank as arrays of amplitudes, or vectors. Thus, it is possible to submit the DFT of an arbitrary sound for pattern matching and thereby use the tool to suggest an orchestration that would "match" (i.e., best reproduce) a reference sound given a set of instruments/notes chosen from the bank. So far, in our tool we have three different processes for data analysis. We have given a simple name to each, which we use in the subsequent discussion.

SVD

Singular value decomposition (SVD) is an advanced method of spectral decomposition used to analyze

an arbitrary target sound in terms of a chosen group of palette instruments. The output of this routine is a set of coefficients, one for each palette sound, representing the amount of the palette sound contained in the target (in the sense discussed in the next section).

This routine is used to get an understanding of the composition of the target in terms of the complete set of palette instruments. By adding more and more sounds with the largest coefficients, the composer can approximate the target sound. Moreover, by examining the sounds with large coefficients, the composer can get new ideas for orchestration. The SVD output gives the weight for all sounds in the palette, ordered from best to worst, so that the composer can choose which is the appropriate one according to musical context.

CHI

Because SVD might return an orchestration which is unplayable, recommending, for example, five simultaneous violin notes for a single violin, CHI is a routine which incorporates performance constraints by taking playable subsets of instruments, notes, and dynamics for the number of palette instruments chosen, combines them, and computes χ^2 (defined in the next section) for each combination. The lower χ^2 is for a particular combination, the better its approximation is to the target.

DOT

The DOT routine computes the inner, or dot product ($\boldsymbol{T} \cdot \boldsymbol{X_j}$) of the target and each sound chosen from the bank, showing individually which sounds have a high match with the target. By ordering palette sounds with highest dot product, we see which individual sounds are most compatible with the target. Although it provides similar information, DOT is rather different from SVD, which returns the coefficients of palette sounds when all are allowed to play together. The DOT routine shows which sound would be the best approximation if played by itself.

Mathematics of the Methods

First, let us fix our notation and rephrase the purpose of our tool in mathematical terms. We start with the DFT of our target sound, represented by the vector $\boldsymbol{T} = \{T_i\} = T(f_i)$. Our sampling resolution $\Delta f = 10.76$ Hz, and the f_i run over 4,096 values, so that $f_{4096} \approx 44.1$ kHz. The averaged DFT of the target sound is created following the same procedure as for the bank sounds.

Our goal is to approximate this target sound, $\boldsymbol{T}$, with a combination of notes and dynamics on a selected set of palette instruments, $\{\boldsymbol{X_j}\}$. j runs over palette instruments, pitches, and dynamics. The DFTs of these will be written variously as $\{X_1(f_i), X_2(f_i), \ldots\} = X_j(f_i) = \boldsymbol{X}_{ij}$, a matrix which we will call the *basis matrix*. (It is a non-orthogonal, incomplete basis for the 4,096-dimensional Fourier space.) These can be selected from the entire bank of sounds. For example, in the first author's work *L'identité voilée*, a clarinet multiphonic $\boldsymbol{T}$ is approximated by various notes and dynamics, $\{\boldsymbol{X_j}\}$ played on piano, violin, and clarinet.

We began by thinking of the goal as a curve-fitting problem. The linear algebra aspects of this problem are quite rich and have also helped to guide our thinking, which has resulted so far in the three routines dicussed next.

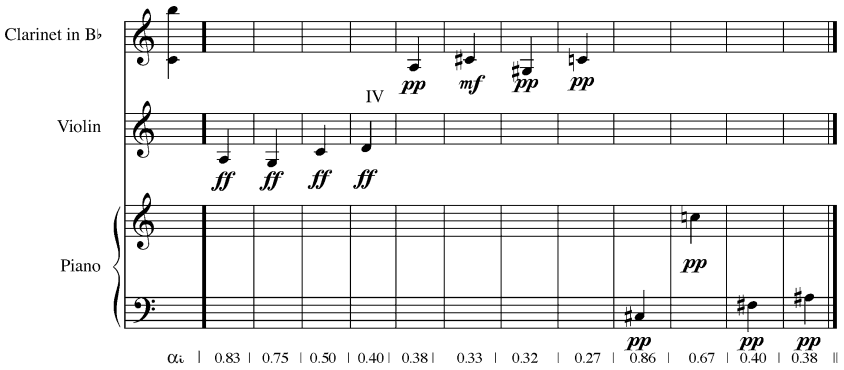
SVD: Generalized Least Squares Fit

The first approach is essentially a generalized "least squares fit" (LSF). From the notes available to the palette instruments chosen, we wish to find the amounts of these notes, whose combined Fourier transform best reproduces the Fourier transform of the target sound, i.e., the "best fit" of the sum of palette instrument notes to the target sound. In the usual least squares problem, one finds the set of coefficients $\{\alpha_j\}$ that makes the polynomial function on the right-hand side the best approximation to the function $y(x)$ on the left:

$$y(x) \sim \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_N x^N = \sum_{j=0}^{N} \alpha_j x^j$$

The function $y(x)$ is known at a discrete set of points $y_i(x_i)$, and the $\{\alpha_j\}$ are best in the sense that they

Figure 2. An SVD example showing the four largest sound contributions for each instrument, and their α_i values, to the clarinet multiphonic target shown in the first column.



minimize, χ^2, the sum of the squares of the differences, $\chi^2 = \sum_i |y_i(x_i) - \sum_j \alpha_j x_i^j|^2$, over points x_i.

In our case, the basis functions are not the polynomials x^n, but instead the DFTs $\{\boldsymbol{X_j}\}$ of the chosen instrument notes. We seek the set of coefficients $\{\alpha_j\}$ that makes the following two functions as close as possible (in the least squares sense of minimal χ^2) at each frequency, f_i:

$$T(f_i) \sim \sum_j \alpha_j X_j(f_i) = \sum_j X_{ij}\alpha_j$$

As outlined in Press et al. (2002), we can find the vector $\boldsymbol{\alpha} = \{\alpha_j\}$ using SVD to solve the problem

$$\mathbf{X}\boldsymbol{\alpha} = \mathbf{T}$$

The solution is

$$\boldsymbol{\alpha} = \boldsymbol{V} \cdot \boldsymbol{w}^{-1} \cdot (\boldsymbol{U}^T \cdot \boldsymbol{T})$$

where $\mathbf{X} = \mathbf{U} \cdot \mathbf{w} \cdot \mathbf{V}^\mathbf{T}$ is the standard SVD of $\mathbf{X}$ (see Press et al. 2002). SVD is ideal for this problem, because for non-square $\mathbf{X}$, as in our case, it gives the "best" solution in the least-squares sense.

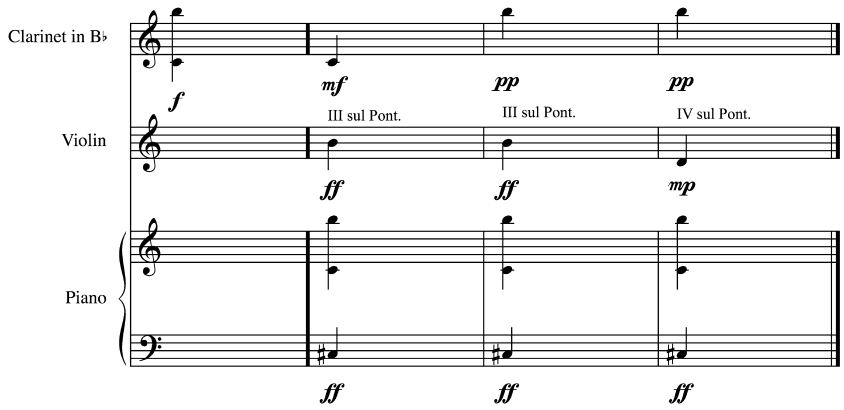
This can be a computationally intense process, as $\mathbf{X}$ has dimensions $M \times N$, where N is 4,096 and M the number of possible instruments, notes, and dynamics available in the palette from which to choose combinations. However, for a "few" instruments (a quintet, say, with their numerous sounds), we can solve for $\boldsymbol{\alpha}$ on a personal computer in a few minutes.

Upon solution, we order the $\{\alpha_j\}$ in descending magnitude so that we can see which instruments, notes, dynamics, and mode of performance make the largest contribution to the approximation of the target sound. It is these large α_j that are the orchestration proposals for the composer. In Figure 2, we show the magnitudes of the α_j for an example $\boldsymbol{T}$ and $\boldsymbol{X_j}$.

This method has advantages and disadvantages. The primary advantage is its comprehensiveness. It returns a coefficient for every sound in the selected basis. Thus, the composer has complete information about the representation of the target using the entire palette chosen for orchestration. Adding more and more notes and instruments with large α_j makes the timbre of the combination closer to the target's. Furthermore, by looking at different combinations of instruments/sounds with the large α_j, the composer can experiment with various orchestrations, each of which captures a different aspect of the target's timbre. This aspect of the tool presents a range of ideas to the composer's creativity.

The drawback is that when orchestrating for a large ensemble, the process becomes numerically challenging in terms of both memory and CPU requirements. Another challenge with this method is that it treats all basis sounds exactly the same: there is no way to specify that the performers can only play one note on their instrument at a time, or that we request up to ten simultaneous notes for the piano. Such restrictions can be done after finding the α_j, by taking the ten largest piano α values, or chosing only the largest clarinet value. However, it would be efficient to have these "performance constraints" built into the tool from the outset. To this end we have explored some other approaches which address these issues.

Figure 3. A CHI example, showing three combinations with low χ^2.



CHI: Performance Constraints

As a simple attempt to address the performance constraints discussed previously, we created arrangements of the palette sounds into combinations, $\boldsymbol{Y_k} = \sum_{j=1}^{N} \boldsymbol{X}_{\{j\}_k}$, of sounds playable by the specified orchestration instruments. For example, if composing for a piano, clarinet, and violin trio, we make the k sets, $\{j\}_k$, of arrangements consisting of all choices of palette sounds composed of, say, eight piano pitches, one clarinet pitch, and one violin pitch (as well as dynamics), in which case $N = 10$ in $\boldsymbol{Y_k}$, and k ranges over the number of possible different choices for these ten notes.

We then compute the χ^2 for each of the permutations, $\boldsymbol{Y_k}$

$$\chi_k^2 = \sum_{f_i} |T(f_i) - Y_k(f_i)|^2$$

for each arrangement. The lower χ_k^2, the better the "fit" of combination $\boldsymbol{Y_k}$ to the target, so that we can choose the best combination, in addition to looking at combinations that are similar (see Figure 3).

Interestingly (and reassuringly), the solutions returned by this method are rather similar to those returned by the complete SVD solution described previously, if we restrict the $\boldsymbol{X_j}$ to sets of playable arrangements. In other words, combinations $\boldsymbol{Y_k} = \sum_{j=1}^{N} \boldsymbol{X}_{\{j\}_k}$, which now have low χ^2, are generally composed of sounds $\boldsymbol{X_j}$ with large α_j in the complete SVD fit to the target.

This method addresses the problem of performance constraints, but it can still be computationally intensive. Even for a small orchestra, the number of possible combinations of playable notes is enormous; thus, this approach is not suitable for a large number of musicians. For example, if we are using five instruments, each of which can play one note at time, from a possible 100 playable notes per instrument, this amounts to 10^{10} combinations for which we must mix and compute χ^2.

DOT: Projection of the Target onto Individual Notes

Our final (in this article) method to approach our goal involves computing the projection of the target onto each of the palette sounds. Our definition of the projection, or inner (dot) product, of $\boldsymbol{T}$ with $\boldsymbol{X_j}$ is

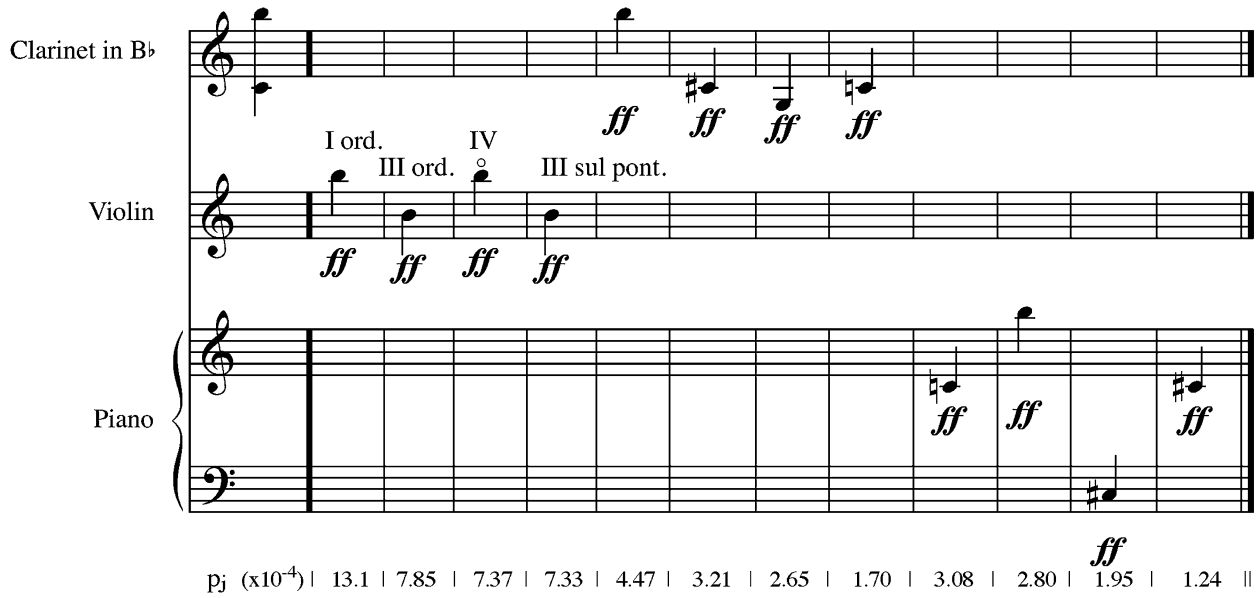
$$p_j = \sum_{f_i} T(f_i) X_j(f_i)$$

It is large when $\boldsymbol{T}$ and $\boldsymbol{X_j}$ are large at the same frequency.

This method is computationally simple because, for a given target, we simply read our palette sounds one by one and compute p_j. We then order the sounds according to the largest projection (see Figure 4).

Although not as complete as the SVD or CHI methods, this approach is simple and fast, and it gives the composer a sense of which instruments

Figure 4. A DOT example, showing the sounds with largest magnitude, p_j, of the inner product of the sound with the clarinet multiphonic target shown first.



and notes in the palette are most comparable to the target (in the vector sense). The composer can then start to examine and combine those with largest projection; or as we do, use a subset with large p_j for combinations of a restricted basis as a start for the CHI method. Thus, DOT provides an efficient preprocessor to eliminate sounds that have very little relation to the target.

If, as in our example, we choose only the ten notes with highest p_j for each of the five instruments, the number of combinations is only 10^5, a huge reduction over the computational requirement (10^{10}) of the CHI example.

An Example: *L'identité voilée*

We now show an example from the first author's *L'identité voilée*, which was composed using our tool. An excerpt from the piece, described herein, will appear on the 2009 *CMJ* DVD and can be found online at http://physics.sci.pacific.edu/~jhetrick/laborch.

In the transition between the first and second sections of *L'identité voilée* for clarinet, violin, and piano, a clarinet multiphonic is presented, and, as it fades out, its resonance is imitated by the entire trio. To determine the specific pitches, dynamics, and performance practices that would best lead to this timbral imitation, the clarinet's multiphonic C4–B5 was analyzed by our tool. Based on the specified three instruments, the tool supplied different solutions. Figure 5 shows one solution proposed by the tool.

On the left-hand side, the solution is presented in score notation (in concert pitch). On the right, the averaged spectrum of the multiphonic on the top is compared with the averaged spectrum of the sound mixture shown in the second measure. To facilitate the comparison, both spectra are displayed over a low C-sharp3. The averaged spectrum of the multiphonic shows that it has strong energy around the 7th, 14th, and 2nd partials of C-sharp3, respectively. The averaged spectrum of the sound mixture matches the strong energy of these partials, and the addition of new partials is rather limited.

The final orchestration of this transition in the piece (an MP3 excerpt is available, as mentioned previously), is shown in Figure 6. The concert experience has demonstrated that the level of imitation was very conclusive.

Figure 5. Left: The clarinet's concert-pitch multiphonic C4-B5, and the pitches, dynamics, and performance practices suggested by the tool to imitate its resonance. Right top: averaged spectrum of the multiphonic; bottom: averaged spectrum of the sound mixture shown in the second measure.

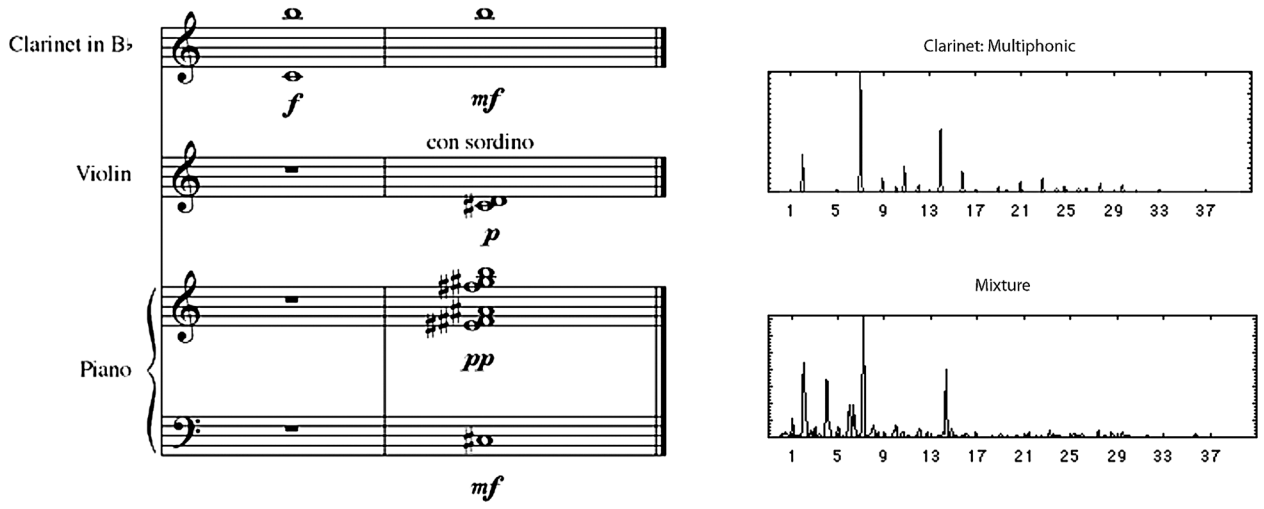


Figure 5

Figure 6. Final score of the transition between first and second second sections of L'identité voilée *by François Rose, following suggestions by the tool.*
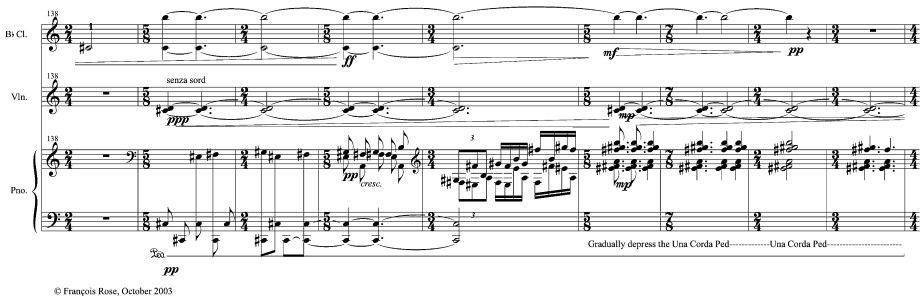


Figure 6

Conclusion and Outlook

We have developed a set of computational routines that provide a composer with information on a palette of orchestration sounds by comparing them to a target sound. These programs have been successfully tested in several compositions by François Rose, where a particular sound is later imitated by the rest of the ensemble. The three modules we have developed so far (SVD, CHI, and DOT) are similar in that they tell us which sounds or orchestrations are closest to the target; however, they each do something different.

The first routine that we explored, SVD, is a general-purpose curve-fitting method. It returns the coefficients of a sum of sounds that "fit" the target better and better (in the least-squares sense) as we include more terms in the sum. Thus, the composer can get a sense of which sounds are most important. In this sense, DOT is very similar; it returns a "match" value for seach palette sound that represents the projection of the target onto the particular palette sound, in a vector sense. A large DOT product means the sounds are similar.

The difference, however, is that DOT gives information for individual sounds in the palette, whereas SVD is more advanced. It includes information about the sounds taken together, as opposed to individually. For example, suppose the target has a strong partial at 440 Hz. If the palette sounds have small but non-zero energy at 440 Hz, they might have low DOT product, individually. Nonetheless,

when taken together, their sum would give a strong 440 Hz signal. Thus, they would appear with relatively high SVD coefficients. Hence, the routines aid the composer in understanding the sounds, but they do not solve the problem of orchestration. By looking at the differences in the sounds recommended by SVD and DOT, and by understanding what the routines are doing algorithmically, the composer gains insight into the orchestration.

The CHI routine stands somewhat apart from the others, as our attempt to include performance constraints into the analysis. It builds performance combinations one by one and compares them to the target in the $\min(\chi^2)$ sense. Because of this combinatorical nature, it is time-consuming. We have found, however, that by "preprocessing" our palette sounds with DOT, we can isolate the important sounds and greatly reduce the number of combinations to be tested.

Our plans for the future are to develop the tool into a stand-alone cross-platform (likely Java-based) tool that would carry out these routines and others in an application with a graphical user interface. We have begun work on this. We would like to add more features to our approach, for example, equalization or other processing of the target sound before analysis. Another feature would be the specification of a particular sound that the composer wants to fix as part of the final orchestration. This can presently be done in the CHI routine by making sure that each combination always has the chosen instrument and pitch.

There are a number of questions we would like to address as well. We have used the norm of the DFT, ignoring the phase information of the complex values. This means that our target and palette functions are always positive semi-definite, whereas the fitting approach that we use is for general functions. So far, this has not been an issue, but we might find some efficiency enhancements if we take the positive nature of the DFTs into consideration. We have also (except for some tests) treated the entire DFT as significant, including small "noisy" values between peaks. If we have a large number of instruments in the palette, this noise could start to influence the fitting routines as much as the peaks. We have not yet seen this, however.

For the latest news on the development of our tool, as well as the musical excerpts mentioned in this article, please visit http://physics.sci.pacific.edu/~jhetrick/laborch.

References

Carpentier, G., et al. 2006. "Imitative and Generative Orchestrations Using Pre-Analyzed Sound Databases." *Proceedings of the Sound and Music Computing Conference.* Marseille, France: 115–122. Available online at mediatheque.ircam.fr/articles/textes/Carpentier06a.

Carpentier, G., et al. 2007. "An Evolutionary Approach to Computer-Aided Orchestration." Available online at mediatheque.ircam.fr/articles/textes/Carpentier07a.

Carpentier, G., D. Tardieu, and X. Rodet. 2007. "Computer-Aided Orchestration Based on Probabilistic Instruments Models and Genetic Exploration." *Proceedings of the 2007 International Computer Music Conference.* San Francisco, California: International Computer Music Association, pp. 188–191.

Hummel, T. A. 2005. "Simulation of Human Voice Timbre by Orchestration of Acoustic Music Instruments." *Proceedings of the 2005 International Computer Music Conference.* Available online at www.thomashummel.net/english/pdf/simulation.pdf.

Press, W. H., et al. 2002. *Numerical Recipes in C, The Art of Scientific Computing*, 2nd ed. Cambridge: Cambridge Univerisity Press.

Psenicka, D. 2003. "SPORCH: An Algorithm for Orchestration Based on Spectral Analyses of Recorded Sounds." *Proceedings of the 2003 International Computer Music Conference.* Available online at http://quod.lib.umich.edu/i/icmc/images/bbp2372.2003.056.pdf.

Rose, F., and J. E. Hetrick. 2005. "Spectral Analysis as a Resource for Contemporary Composition Technique." *Proceedings of the Conference on Interdisciplinary Musicology.* Montréal: Université de Montréal, pp. 112–113.

Rose, F., and J. E. Hetrick. 2007. "L'analyse spectrale comme ressource à la technique orchestrale contemporaine Cahiers de la SQRM." *Société Québécoise de Recherche en Musique* 9(1–2):63–68.